PHYSICS-INFORMED MODEL DISCOVERY FOR SUBGRID-SCALE TURBULENCE IN RAYLEIGH-BÉNARD CONVECTION

A Thesis Proposal

by

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I. Introduction

Turbulent flows are among the most computationally challenging problems in fluid dynamics, requiring resolution down to the smallest spatial and temporal scales. Thus, the scale range spans multiple orders of magnitude in size. Direct Numerical Simulation (DNS), which resolves all scales down to the smallest one, provides the highest fidelity representation but demands prohibitive computational resources that scale with the Reynolds number as $Re^{9/4}$ [1]. This cost renders DNS impractical for most real-world high-Reynolds number applications [2], as those range in between $Re = 10^6$ and $Re = 10^{12}$.

The common solution to this issue is to only solve for the largest scales and model the effect of the small unresolved ones. The fundamental challenge with this approach is the closure problem, i.e. the additional term describing the effect of the unresolved scales on the resolved ones. The closure term cannot be accurately parameterized in terms of the large scales, as it contains information about the subgrid scales (SGS). Since subgrid scales cannot be resolved on the computational grid, their effect must be modeled [3].

Traditional large eddy simulation (LES) models, like the dynamic Smagorinsky model, rely on phenomenological assumptions. While they are useful, there are two major issues with these models. Firstly, they are only applicable to situations where the specific assumptions tied to the model hold. Secondly, it is common to clip backscatter (energy transfer from small to large scales) to ensure numerical stability. This practice simplifies the models to the extent that they lose accuracy in the long-term [4].

Recent advances in machine learning have catalyzed a shift toward data-driven turbulence modeling. These approaches leverage high-fidelity DNS or experimental data to learn closure models directly, discovering mathematical relationships from data while respecting physical constraints [5]. However, the most popular modeling approaches that rely on deep learning suffer from the common problems of low interpretability, overfitting, high requirements for the amount of training data, and especially poor out-of-sample performance.

The core idea of this proposal is to generate high-fidelity DNS data and then use machine learning, specifically the SPIDER (Sparse Physics-Informed Discovery of Empirical Relations) framework [6], to discover a symbolic, equivariant, and accurate closure model directly from the data. This approach would eliminate the trade-offs of both traditional first-principles modeling and deep-learning-based modeling, as it requires no restrictive phenomenological assumptions and yields a directly interpretable mathematical formulation.

II. Methodology

This proposal focuses on 2D Rayleigh-Bénard convection (RBC), which describes a buoyancy-driven flow in a fluid layer heated from below and cooled from above, traditionally with rigid boundaries on the top and bottom. Its well-defined geometry and boundary conditions, combined

with complex multiscale dynamics, make it an ideal testbed for evaluating model discovery algorithms across varying complexity levels [7]. It is governed by three equations describing mass, momentum, and heat transport, respectively:

$$\text{Mass:} \qquad \partial_i u_i = 0 \tag{1}$$

$$\text{Momentum:} \qquad \partial_t u_i + u_j \partial_j u_i = -\partial_i p + \nu\, \partial_{jj} u_i + \bar{b}\delta_{i3} \tag{2}$$

$$\text{Heat:} \qquad \partial_t b + u_j \partial_j b = \kappa\, \partial_{jj} b \tag{3}$$

The computational domain has no-slip, fixed-temperature boundary conditions at the top and bottom and periodic (repeating) boundary conditions on the sides.

The flow is characterized by the Rayleigh number (Ra), which measures the driving strength relative to dissipative effects. As Ra increases, the flow progresses through distinct regimes: from stationary ($Ra \leq 1708$), through laminar convection rolls ($Ra < 1e7$) to fully developed turbulence ($Ra > 1e7$). We will focus exclusively on turbulent flows. The RBC's relevance extends to atmospheric and oceanic processes, geophysics, and various industrial energy systems [2]. The system's configurability, definition, and wide range of applications make it a perfect fit for studying problems in CFD. In the proposed study, we focus on Rayleigh numbers $Ra = 1e8; Ra = 1e10; Ra = 1e12$ to systematically stress-test the proposed solutions [8].

Spatial filtering yields equations for the large-scale (resolved) velocity, pressure and buoyancy (scaled temperature) fields, denoted by the bar:

$$\partial_i \bar{u}_i = 0 \tag{4}$$

$$\partial_t \bar{u}_i + \bar{u}_j \partial_j \bar{u}_i = -\partial_i \bar{p} + \nu\, \partial_{jj} \bar{u}_i + \bar{b}\delta_{i3} - \partial_j \tau_{ij} \tag{5}$$

$$\partial_t \bar{b} + \bar{u}_j \partial_j \bar{b} = \kappa\, \partial_{jj} \bar{b} - \partial_j \phi_j \tag{6}$$

where τ_{ij} and ϕ_j are the closure terms given by

$$\text{Momentum equation closure:} \qquad \tau_{ij} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j \tag{7}$$

$$\text{Buyancy equation closure:} \qquad \phi_j = \overline{u_i b} - \bar{u}_i \bar{b} \tag{8}$$

Both terms arise from the nonlinear advection terms and must be modeled, traditionally as functions of the resolved fields [3]. Many phenomenological LES models including the dynamic Smagorinsky model (DSM), are predicated on the concept of scalar eddy viscosity. They assume that dissipation is determined by the resolved strain rate and often require clipping for numerical stability, making the model purely dissipative [9]. These assumptions break down in buoyancy-driven flows, where the physics are inherently direction-dependent and involve complex energy backscatter that heuristic models completely omit [4]. There is currently no rigorous methodology for SGS closure modeling that respects the unique physics of thermal convection without relying on restrictive phenomenological assumptions.

This proposal aims to solve this problem by establishing a principled, assumption-free framework for discovering SGS closures directly from data. We do this by leveraging The Sparse Physics-Informed Discovery of Empirical Relations (SPIDER), which is a powerful framework for extracting symbolic equations from data. It combines three key elements: physical constraints imposed by symmetry, a weak formulation of differential equations, and sparse regression [10].

SPIDER incorporates physical principles like translational or rotational invariance directly into the library construction process, constraining the search space and ensuring physical consistency. This will ensure that the discovered model inherits the symmetries of the governing equations. Its weak formulation provides exceptional robustness to noise and discretization, allowing SPIDER to extract accurate relations from highly turbulent datasets where traditional methods fail [5]. The framework has demonstrated remarkable success across diverse systems, recovering governing equations in a range of situations [6]. Recently, it has also been used on SGS model discovery in turbulent 2D situations [11]; my work will build on this study.

By leveraging high-fidelity DNS data and sparse regression (SPIDER), we aim to uncover the fundamental mathematical structure of the subgrid stresses and heat fluxes. Unlike ”black box” neural networks, this approach yields explicit, interpretable algebraic expressions, allowing us to directly link the discovered terms to the underlying physics of scaling and scale interactions [6]. The ultimate goal is to validate these discovered closures *a posteriori*, delivering a numerically stable, physically consistent model that outperforms existing subgrid-scale models and establish a new standard for LES of turbulent convection.

To assess whether this interpretable subgrid-scale model truly captures the governing physics, we will subject it to the full suite of *a priori* and *a posteriori* evaluations used in turbulent Rayleigh-Bénard convection. In the *a priori* setting, we compare the discovered stresses and heat fluxes against the same quantities computed from filtered DNS solutions. We will then perform *a posteriori* tests by running standalone LES and verifying that it reproduces the essential physics of convection—accurate scaling between Nusselt and Rayleigh numbers, realistic boundary-layer structure, power spectra, and plume dynamics——while maintaining stability. This progression from targeted *a priori* validation to full-system *a posteriori* performance provides a coherent pathway for demonstrating that SPIDER-derived models are not only interpretable, but also predictive and robust in practical LES applications.

This research direction has the potential to advance the fundamental understanding of turbulent heat transfer by bridging classical turbulence theory with modern, physics-informed machine learning. Demonstrating that frameworks like SPIDER can recover physically accurate and numerically stable SGS closures would be a significant methodological advance in modeling of multiscale phenomena [10].

III. Aims and Approaches

The primary aims of this research project are structured into three distinct phases:

1) To create high-quality datasets from computer simulations of two-dimensional Rayleigh-Bénard convection (RBC) using DEDALUS, a high-accuracy spectral solver framework, which uses the Boussinesq approximation. The fluid's velocity, pressure, buoyancy, and vorticity fields across a range of Rayleigh and Reynolds numbers will be recorded. DNS 2D Rayleigh-Bénard convection will span Rayleigh numbers $Ra = 10^8, 10^{10}, 10^{12}$. SPIDER will be used to validate the fidelity of the data.
2) To use spatial filtering to coarse-grain the DNS data and derive the subgrid-scale (SGS) stress and heat flux terms. We will use the coarse-grained data and SPIDER to recover a principled, accurate and interpretable model for the SGS closure directly from the data.
3) To validate and interpret the discovered closure models. We will accomplish this by thoroughly analyzing errors, checking if the models make sense physically, and testing their robustness and performance across different scales and flow conditions. We will validate the *a priori* and *a posteriori* performance of the discovered model and compare it to the currently used phenomenological models and DNS data across different Rayleigh numbers.

IV. Conclusion

This proposal addresses the fundamental computational bottleneck in simulating turbulent flows: the prohibitive cost of resolving multiscale interactions. By focusing on Rayleigh-Bénard convection, a canonical system for buoyancy-driven turbulence, this research seeks to bridge the gap between the high fidelity of DNS and the computational efficiency of LES required for practical applications.

This work innovates in this direction by integrating of the SPIDER framework into SGS modeling. Unlike standard data-driven approaches that utilize opaque neural networks to approximate SGS terms, SPIDER leverages sparse regression to discover interpretable, explicit algebraic closure equations. This distinction is critical, as it ensures that the resulting models not only achieve high predictive accuracy but also respect fundamental physical symmetries, significantly enhancing their generalizability and numerical stability compared to phenomenological LES models like the Smagorinsky model.

The successful execution of this research will offer a profound and broad impact on both theoretical and applied physics. In the domain of computational fluid dynamics, establishing a robust workflow for discovering interpretable SGS closures would set a new standard for LES, reducing the reliance on manual parameter tuning. Practically, improved modeling of turbulent thermal convection has direct implications for geophysics and climatology, where accurate parameterization of heat flux is essential for reducing uncertainty in climate projections [2]. Ultimately, this work aims to demonstrate that machine learning can serve not just as a black-box approximator, but as a tool for understanding physical systems.

V. Final Project Timeline

Fall 2025 (Foundation Stage)

- [High] Install and configure Dedalus on cluster workstation environment (by Sept. 30, 2025).
- [High] Set up Rayleigh–Bénard convection simulations in Dedalus (by Oct. 15, 2025).
- [High] Generate initial high-resolution datasets across parameter ranges (by Oct. 31, 2025).
- [Medium] Interpolate Chebyshev-basis data onto uniform grids (by Nov. 15, 2025).

Late Fall 2025 – Spring 2026 (Model Discovery Stage)

- [High] Apply Gaussian filtering to DNS data (Dec. 2025).
- [High] Infer closure term from filtered fields (Dec. 2025).
- [High] Compare the inferred closure with the one obtained using moment expansion (Jan. 2026).
- [Medium] Document results and progress in weekly meetings (Ongoing, weekly).

Spring 2026 (Communication Stage)

- [High] Conduct *a priori* evaluation of discovered closures and quantitatively compare models to DNS data (Mar. 2026).
- [Medium] Comparing predictions of the original and coarse-grained models (by Mar. 15, 2026).
- [Medium] Analyze scaling of coarse-grained model accuracy with filter scale (by Mar. 31, 2026).
- [Medium] Prepare poster summarizing model discovery results (by Apr. 10, 2026).
- [Medium] Present at Spring Symposium (Apr. 17, 2026).

Summer 2026 (Evaluation Stage)

- [Medium] Identify and document limitations of models (Jul. 2026).
- [Low] Refine models based on evaluation results (Aug. 2026).

Long-term (2026+)

- [Low] Draft and revise manuscript for peer-reviewed journal (Fall 2026–Spring 2027).
- [Low] Submit final paper for publication (Spring 2027).

Priority Key

[High] = High Priority (critical, prerequisite for later tasks)
[Medium] = Medium Priority (important for progress and communication)
[Low] = Low Priority (long-term refinement and dissemination)

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